Tutorial

property features and prices

Area	Bedrooms	Age	Distance	price
1500	3	5	10	250
1800	3	3	8	300
1200	2	10	15	180
2000	4	2	5	350
1600	3	7	12	240

(1) write the ridge regression cost function

$$y_n = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$J(\theta) = \frac{1}{2m} \Sigma (y^n - y)^2 + \lambda(\theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2)$$

Given $\theta = 30$, $\theta_1 = 2$, $\theta_2 = 0.3$, $\theta_3 = 0.4$, $\theta_4 = 1$

$\alpha = 0.001$, $\lambda = 0.8$, $n = 10$

$$y_1^n = 30 + 2(1500) + 0.3(3) + 0.4(5) + 1(10)$$

$$= 30 + 3000 + 0.9 + 2 + 10$$

$$y_1^n = 8042.9$$

$$\frac{1}{2n} \Sigma (\hat{y} - y)^2 = \frac{227868082.1}{10}$$

$$= 22786808.21$$

$$J(\theta) = 22786808.21 + 0.8(25198472.4)$$

$$= 22786808.21 + 20158777.92$$

$$J(\theta) = 42945586.13$$

$\theta_3 = 0.4 - 0.001\,[15018.94 + 2(0.8)(0.4)]$

$\theta_3 = -14.61958$

$\theta_4 = \theta_4 - \alpha\left(\frac{1}{n}\sum(\hat{y}-y)x_4 + 2\lambda\theta_4\right)$

$\sum(\hat{y}-y)x_4 = 2792.9(180) + 3340.13(8) + 2269.6(15)$
$+ 3687(5) + 3005.7(12) = 143197.2$

$\frac{1}{n}\sum(\hat{y}-y)x_4 = \frac{143197.2}{5} = 28{,}639.44$

$\theta_4 = 1 - 0.001\,(28{,}639.44 + 2(0.8)(1))$

$= 1 - 0.001\,(28641.04)$

$\boxed{\theta_4 = -27.64104}$

$\theta_0 = 26.98 \quad \theta_1 = -5019.63 \quad \theta_2 = -9.04$

$\theta_3 = -14.61 \quad \theta_4 = -27.64$

Ridge function

$J(\theta) = \frac{1}{2n}\sum(\hat{y}-y)^2 + \lambda\,(\theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2)$

Here $n=5 \quad \lambda = 0.8$

$\sum(\hat{y}-y)^2 = 227868082.1$

$W_{new} = W_{old} + \text{learningrate} \cdot y \cdot x$

$= [0, 0, 0, 0] + 0.01 \cdot (+1) [50, 30, 180, 90]$

$= [0.5, 0.3, 1.8, 0.9]$

<u>S-2</u> $[28, 22, 100, 70]$ $y = -1$

$f(x) = (0.5)(28) + (0.3)(22) + (1.8)100 + (0.9)(70) = -263.6$

$= (-1) \cdot f(x) \geq 1 \rightarrow$ correct

$y f(x) \geq 0 = -263.6 \rightarrow$ not

need update weights

Updated weights

$w = (0.5, 0.3, 1.8, 0.9) + 0.01 \times (-1)(28, 22, 100, 70)$

$\Rightarrow (0.22, 0.08, 0.8, 0.2)$

(3) $x_3 = (55, 32, 190, 95)$

$w = (0.22, 0.08, 0.8, 0.2)$

$f(x) = 185.66$

$y f(x) \Rightarrow 185.66 > 1$

(4) $w = (0.22, 0.08, 0.8, 0.2)$

$x_4 = (25, 20, 95, 68)$

$f(x) = 96.7$

$y \times f(x) = (-1) 96.7$

$= -96.7 < 1$